

The Computational Mesoscale

Mechanistic Inductive Bias Between Neural Execution and Model Behavior

Ivan GAN

ivan@ivan.icu www.ivan.icu github.com/ivanicu

Abstract

Architecture science predicts what neural networks can compute and how efficiently they learn. Mechanistic interpretability explains particular trained instances. Missing between them is a theory that predicts *how a successful computation will be internally organized before the model's internals are inspected*. We introduce that organization as a dependent variable of architecture science. Two behaviorally matched models may preserve different intermediate distinctions, place them in different carriers, realize the same abstract operation through different physical processes, and fail at different interfaces. We call the architecture- and training-conditioned distribution over such implementation-equivalence classes *mechanistic inductive bias*.

The paper develops a theory-building framework around one claim: architecture does not determine an internal ontology; it changes the relative accessibility, stability, transport, and failure scaling of candidate mechanisms. We model architecture as a typed causal syntax of paths, sharing, routing, cuts, memory, position, execution order, and resources. The forward problem predicts a mechanism distribution; the inverse problem recovers admissible implementation worlds under held-out interventions; the edit problem tests whether a controlled syntax change moves that distribution in the predicted direction. Typed structural results for reachability, robust cut capacity, and quotient composition yield a carrier-cut separation: independently queryable bindings behind a fixed robust cut require exponentially growing distinguishability, whereas an extensive family of addressable carriers relocates pressure from compression to selection. This separation organizes falsifiable predictions for MLPs, CNNs, recurrent/state-space models, and Transformers, including a stage-resolved Transformer account of storage, addressing, transformation, publication, consumption, and composition. The framework is rejected if syntax does not add held-out predictive information about mechanisms, failures, or architecture-edit responses beyond task, training, resource, and behavioral baselines.

Keywords: computational mesoscale; neural architecture; mechanistic inductive bias; virtual computation; mechanistic interpretability; model behavior

1 Introduction

1.1 The dependent variable that architecture science is missing

Consider two models that answer the same associative-retrieval query with the same accuracy. One preserves source-local key-value records and later selects one by content. The other compresses the assignments through a recurrent cut and reconstructs the value at query time. Standard architecture comparison can report that both models express and learn the task. It cannot name the variable on which they differ, predict whether increasing load will cause collision or misaddressing, or state which architecture edit should move one implementation toward the other.

The missing dependent variable is *internal computational organization*: which intermediate distinctions are preserved, where they are physically carried, which processes implement abstract transformations, how those processes compose, and where failure first enters the causal protocol. Behavioral equivalence underdetermines all of these. A task can be solved by addressable records, compressed state, a scan, a shortcut, or an input-conditioned mixture. [Figure 1](#) shows why the difference is scientifically consequential even before it changes accuracy.

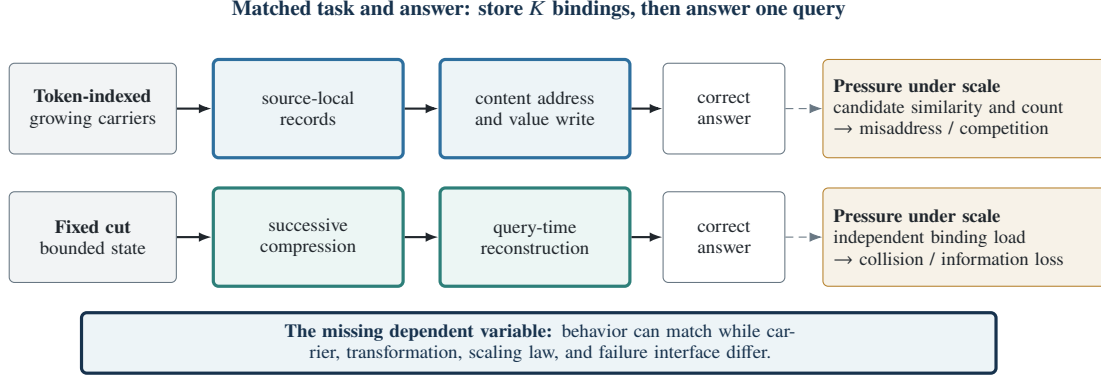


Figure 1: Same function, different computation. A growing addressable carrier family and a fixed recurrent cut can produce the same answer at low load while predicting different internal organizations and different failure scaling. AVC treats that difference as a dependent variable of architecture science.

Architecture research already studies function classes, optimization, scaling, efficiency, and behavioral inductive bias. The proposal here is a second-order inductive bias. A fully specified causal syntax and training environment induce a distribution over implementation organizations, evaluated at a declared equivalence level. We call this distribution *mechanistic inductive bias*.

Theory claim. *Neural architectures bias not only which functions are learned, but which mechanisms realize them. Architecture supplies a causal syntax; training and context select an internal organization; held-out causal experiments recover and test that organization.*

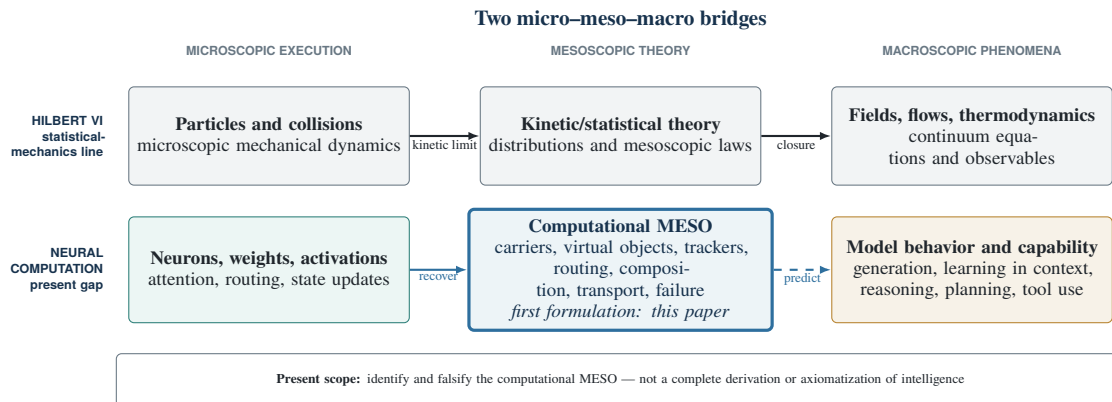
The statement is not that an architecture brand carries a fixed ontology. Rather, syntax changes the probability, stability, transport, and characteristic failure of candidate implementations. It predicts how controlled architecture edits should move mechanism distributions, not merely how a trained model can be redescribed after inspection.

1.2 Paper identity, scope, and contributions

This is a *theory-building framework paper*: it introduces a scientific object, derives a first architecture-conditioned separation, and specifies how the theory can be estimated and falsified. It is neither a report of a completed cross-architecture benchmark nor a manifesto that treats future mathematics as present evidence. Its purpose is to make internal organization a predictive target of architecture science. The complete programme extends much further—through mechanism formation, world switching, cross-model transport, reconstruction, and architecture design—but this paper develops only the formal and experimental layer required to make those later questions scientific.

A Hilbertian programme for neural computation, bounded at the first bridge. Hilbert’s sixth problem called for axiomatic treatment of those physical sciences already receptive to mathematical formulation. One canonical line of that programme seeks controlled passages from microscopic mechanics, through kinetic or statistical descriptions, to continuum laws (Hilbert, 1902; Gorban, 2018). AVC invokes this *research architecture*, not the subject, maturity, or achieved status of the physical programme. The relevant lesson is that stable macroscopic regularities need not be read directly from individual particles: an intermediate vocabulary and its laws can be scientifically indispensable.

Neural systems present the same open shape. At the microscopic end, every parameter, activation, routed message, and state update is physically executable and increasingly observable. At the macroscopic end, trained models exhibit organized behavior—generation, retrieval, in-context adaptation, reasoning, planning, and tool use. Whatever philosophical threshold one adopts for the word *intelligence*, these structured capabilities are empirical phenomena that require explanation. What is missing is a disciplined causal level between them.



The parallel is deliberately asymmetric. In statistical mechanics, candidate micro-, meso-, and macro-theories can be named before the limiting relation is proved. In a trained neural system, the intermediate objects and laws are themselves unknown. The neural problem is therefore *inverse as well as axiomatic*: recover a stable intermediate theory from physical execution, then retain it only if it predicts held-out interventions, transport, composition, reconstruction, and failure. The computational MESO is not a particular neural layer or a convenient summary; it is a candidate causal level of description that compresses microscopic execution while preserving the distinctions needed to explain macroscopic behavior. This paper offers a first formulation and a license test for that level, not a derivation of intelligence in full. Later closures would have to connect computational organization to structural semantics, grounding, agent–environment loops, and alignment under the same standards. Only then could the resulting stack become a common formal spine for model interpretability and alignment.

The paper makes four primary contributions.

1. It defines *mechanistic inductive bias*: the architecture- and training-conditioned distribution over internal implementation classes, and joins its forward prediction, inverse recovery, and edit-based causal test.
2. It gives a typed structural scaffold and a carrier–cut separation that connects architecture-level memory organization to distinct scaling pressures and failure interfaces.
3. It derives falsifiable predictions for four core architecture families and uses a worked Transformer protocol to show how carrier, address, transformation, publication, consumer, competition, and composition failures become distinguishable hypotheses.
4. It supplies the evidence firewall, structural holdouts, architecture edits, and phased empirical protocol required to accept or reject the theory.

1.3 Relation to neighboring programmes

AVC does not replace expressivity, inductive-bias, representation-comparison, causal-abstraction, or mechanistic-interpretability research. It changes their junction. Classical architecture theory constrains possible functions and resources; empirical inductive-bias work observes behavior and representation geometry; mechanistic interpretability recovers local objects and processes; causal abstraction tests relations between low- and high-level models (Cohen and Welling, 2016; Ravanbakhsh, 2020; Geiger et al., 2023; Pearl, 2009). AVC asks whether architecture syntax predicts the *distribution* of recovered computational organizations and whether syntax edits causally move that distribution.

Table 1: AVC at the junction of neighboring research programmes. The rows are complementary, not claims of exclusive territory.

Programme	Primary question/output	Gap addressed by AVC
Expressivity and complexity	Which functions or languages can the architecture realize under resource assumptions?	Does not determine which successful internal organization training selects
Behavioral inductive bias	Which functions, cues, or generalizations are learned more readily?	Usually treats internal mechanism as latent or secondary
Representation comparison	Which geometries, features, or alignments appear across models?	Similarity does not establish native transformations, composition, or failure interfaces
Mechanistic interpretability	Which internal variables and circuits causally contribute in a trained model?	Usually begins after architecture and does not estimate an architecture-conditioned mechanism distribution
Causal abstraction	Does a low-level system faithfully implement a proposed high-level causal model?	Often requires the high-level variables or alignment family to be supplied
AVC	Which implementation organizations are made relatively likely by syntax and training, recoverable under contracts, and shifted by architecture edits?	Unifies forward mechanism prediction, inverse recovery, and edit-based falsification

1.4 Roadmap

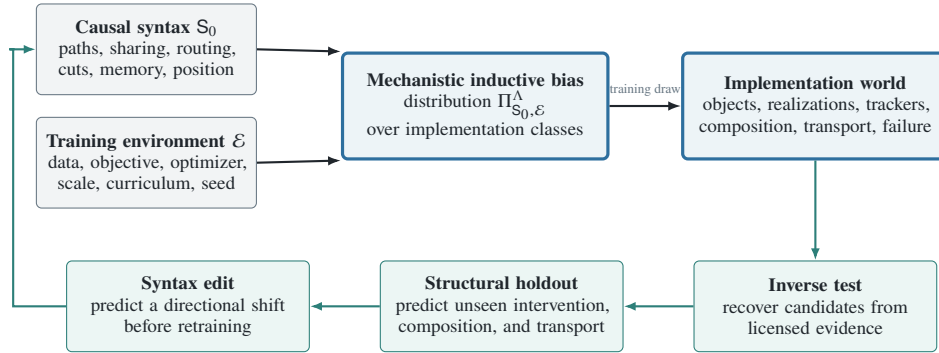


Figure 2: The theory in one view. The forward model predicts a distribution over implementation organizations. The inverse, holdout, and edit tests then ask whether that prediction survives evidence it was not fitted to. The loop is scientific only when the syntax edit is declared before the new model is trained and inspected.

2 Architecture as causal syntax and virtual computational organization

2.1 The missing mesoscale

The word *virtual* is used in the engineering sense of a virtual machine or protocol layer, not to imply that the organization is imaginary or disembodied. A virtual computational organization is a causally stable description above individual coordinates and below end-to-end behavior. It specifies which abstract distinctions downstream computation preserves, which physical states realize them, which processes transform them, and which compositions and transports remain valid under declared interventions.

The target sits between microscopic execution and macroscopic behavior:

$$\underbrace{\text{weights, activations, routes}}_{\text{physical execution}} \longrightarrow \underbrace{\text{objects, trackers, composition}}_{\text{virtual computation}} \longrightarrow \underbrace{\text{task behavior and failure}}_{\text{external consequence}}$$

Microscopic traces do not arrive pre-partitioned into computational objects, while task structure does not uniquely identify the model’s implementation. Interventions constrain the middle layer from below; task counterfactuals constrain it from above.

2.2 Notation and generative objects

The notation deliberately separates the external task, trained physical model, measurement method, evidence splits, and recovered implementation claim.

Table 2: Disjoint symbol families for task, model, method, data, and recovery objects.

Symbol	Kind	Meaning
$\mathcal{T}$	task specification	Externally defined abstract states, operations, observations, and task interventions
S_0	model syntax	Nominal typed graph, execution order, sharing, exact symmetries, routing class, cuts, resources
θ	model parameter	A draw from stochastic training
$S_\theta^{\text{op}}(x, c)$	execution trace	Exact operational dependencies for one run, before empirical thresholding
$\mathfrak{F}$	experimental frame	Context scope, admissible interventions, native/readout outcomes, graph granularity
$\mathcal{W}_{\theta, z}^{\text{impl}}(\mathfrak{F}, \varepsilon)$	implementation world	Stable objects and typed trackers for mechanism state z at tolerance ε
$\mathfrak{D}_m$	method syntax	Target, observables, interventions, alignment class, aggregation, scope, and cost
$\mathcal{D}_{\text{fit}}, \mathcal{D}_{\text{sel}}, \mathcal{D}_{\text{eval}}, \mathcal{D}_{\text{hold}}$	data	Disjoint method fitting, selection, evaluation, and structural holdout splits
$\widehat{\mathfrak{M}}_{\varepsilon, \delta}$	recovery output	Candidate worlds not rejected at tolerance ε and uncertainty δ

2.2.1 Generation layer

Definition 1 (Nominal model syntax). A model specification has nominal syntax

$$S_0 = (\text{Ty}, \mathcal{G}_0, <, \sim_{\text{par}}, G_{\text{exact}}, \rho_0, \text{Cut}_0, \text{Res}_0).$$

Ty lists physical state types; $\mathcal{G}_0$ is the potential typed dependency graph; $<$ is execution order; $\sim_{\text{par}}$ is parameter sharing; G_{exact} contains only exact group actions; ρ_0 is the routing class; Cut_0 lists graph cuts at a declared node granularity; and Res_0 records finite-precision, bandwidth, memory, and compute assumptions. Approximate symmetries are not elements of G_{exact} ; they are metric claims with a defect and tolerance.

Architecture labels are not sufficient syntax. A “Transformer” specification must disclose mask, position system, normalization, residual organization, attention sparsity, cache semantics, precision, and any recurrence or expert routing. A “CNN” must disclose padding, stride, pooling, dilation, boundary convention, and readout. Cuts and resources are consequently frame- and granularity-indexed, not brand-level invariants.

Let $\mathcal{E}$ denote the data distribution, objective, optimizer, curriculum, regularization, initialization law, and implementation regime. Training is explicitly stochastic:

$$\xi \sim p_{\Xi}, \quad \theta \sim \text{Train}(S_0, \mathcal{E}, \xi).$$

The forward problem predicts a distribution over mechanism states and implementation worlds, not a unique world:

$$p(z, \mathcal{W}_{\theta, z}^{\text{impl}} | S_0, \mathcal{E}, x, c).$$

2.3 Operational execution and empirical effective syntax

Definition 2 (Operational execution syntax). $S_\theta^{\text{op}}(x, c)$ is the fully resolved typed execution graph for one run: instantiated gates, selected experts, attention support before effect thresholding, residual and skip edges, normalization dependencies, shared randomness, caches, and sampler state. It is a property of the structural equations and runtime state, not of an interpretability method.

Operational support is not empirical causal importance. Attention weights, gate values, or nonzero Jacobian entries can coexist with zero relevant downstream effect. We therefore reserve

$$\widehat{S}_\theta^{m, \mathfrak{F}, \mathfrak{C}_{\text{do}}, \varepsilon}(x, c)$$

for an *empirically recovered effective syntax*: edges or coalitions whose declared causal estimand exceeds tolerance ε under method m , frame $\mathfrak{F}$, and intervention contract $\mathfrak{C}_{\text{do}}$. This indexing removes the circular fiction that “causally active” edges are method-free model attributes.

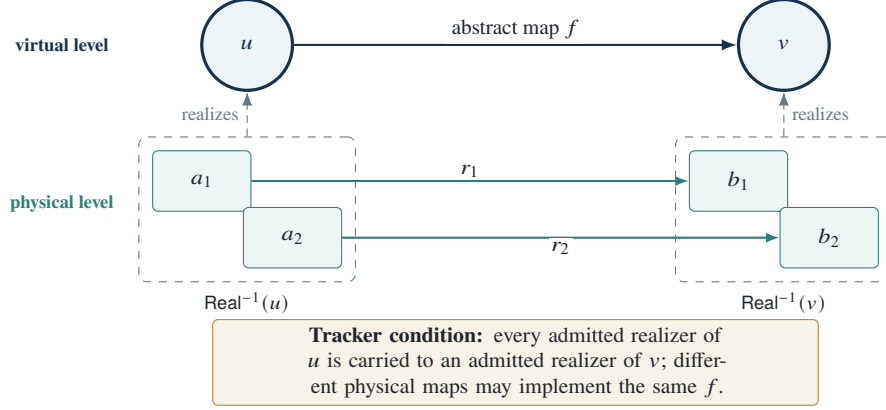


Figure 3: Anatomy of an implementation world. Virtual objects are equivalence classes supported by one or more physical realizers. A physical process is a tracker only when it acts uniformly across the declared realization family and descends to the abstract map. This separates multiple realization from a single preferred coordinate or circuit.

2.4 Task, implementation, and recovery

Definition 3 (Task specification). The external task is

$$\mathcal{T} = (\mathcal{U}_{\mathcal{T}}, \mathcal{F}_{\mathcal{T}}, \mathcal{O}_{\mathcal{T}}, \mathcal{I}_{\mathcal{T}}, p_{\mathcal{T}}),$$

where $\mathcal{U}_{\mathcal{T}}$ is an abstract state set, $\mathcal{F}_{\mathcal{T}}$ a typed family of task operations, $\mathcal{O}_{\mathcal{T}}$ task observations, $\mathcal{I}_{\mathcal{T}}$ permitted task interventions, and $p_{\mathcal{T}}$ the generator. $\mathcal{T}$ is ground truth about the *task*; it is not ground truth about a trained model's internal mechanism.

Definition 4 (Typed implementation world). For parameters θ , coherent mechanism state z , frame $\mathfrak{F}$, and tolerance ε , an implementation world is

$$\mathcal{W}_{\theta, z}^{\text{impl}}(\mathfrak{F}, \varepsilon) = (\mathbf{U}_z, \mathbf{X}_z, \{\sim_{\tau, z}\}_{\tau \in \text{Ty}}, \text{Real}_z, \text{Trk}_z, \text{Comp}_z, \text{Trans}_z, \Gamma_z).$$

$\mathbf{U}_z$ is a typed family of virtual objects; $\mathbf{X}_z$ the scoped physical carriers; $\sim_{\tau, z}$ the intervention-and-reader equivalence on carriers of type τ ; $\text{Real}_z \subseteq \mathbf{X}_z \times \mathbf{U}_z$ the realization relation; Trk_z a typed family of physical maps that preserve relevant equivalences; Comp_z the declared abstract/physical composition tests; Trans_z partial correspondences across contexts, positions, times, seeds, or models; and Γ_z records scope, approximation defects, resource costs, and unresolved alternatives.

Definition 5 (Recovered candidate set). After method fitting and selection are frozen, recovery returns

$$\widehat{\mathfrak{M}}_{\varepsilon, \delta} = \text{Recover}(\mathcal{D}_{\text{eval}}, \mathcal{D}_{\text{hold}} \mid \mathbf{S}_0, \mathcal{E}, \mathfrak{Q}_m, \mathfrak{F}, \mathfrak{C}_{\text{do}}, \mathfrak{C}_{\text{id}}),$$

the set or posterior of implementation worlds not rejected at error tolerance ε and uncertainty level δ . Recovery is evaluated by held-out interventional prediction, compression under a declared complexity class, typed composition, transport, and constructive tests. Metaphysical uniqueness is not an output.

The three objects are connected but not identified:

$$\mathcal{T} \xrightarrow[\text{training}]{\mathbf{S}_0, \mathcal{E}, \xi} \mathcal{W}_{\theta, z}^{\text{impl}} \xrightarrow[\text{measurement}]{\mathfrak{Q}_m, \mathfrak{F}, \mathfrak{C}_{\text{do}}} \mathcal{D}_{\text{eval}} \xrightarrow[\text{recovery}]{\mathfrak{C}_{\text{id}}} \widehat{\mathfrak{M}}_{\varepsilon, \delta}.$$

A task operation can have multiple implementation trackers; a trained model can use a shortcut not isomorphic to the task decomposition; and several candidate implementation worlds can predict the same observed evidence.

2.5 Cross-input coherence and world switching

Allowing an arbitrary world for every input makes the theory unfalsifiable. We therefore introduce a mechanism state

$$z \sim p_\psi(z \mid x, c, \theta), \quad z \in \mathcal{Z},$$

where $\mathcal{Z}$ and its granularity are fixed before final evaluation by one of three declared rules: a planted label, a preregistered finite family, or a complexity-penalized partition selected on $\mathcal{D}_{\text{sel}}$. Each z has a stable world on a scope $\Omega_z \subseteq \mathcal{X} \times \mathcal{C}$. New states cannot be created after inspecting $\mathcal{D}_{\text{hold}}$.

For a declared context arrow $\gamma : c \rightarrow c'$, transport is a partial correspondence

$$\mathbf{X}_{z,c} \xleftarrow{\pi_c} K_\gamma \xrightarrow{\pi_{c'}} \mathbf{X}_{z',c'},$$

not necessarily an invertible conjugacy. Identity and cocycle defects are evaluated on the common domain:

$$\Delta_{\text{id}}(\gamma) = d(\text{Trans}_{\text{id}_c}, \text{id}), \quad \Delta_{\text{cyc}}(\gamma_2, \gamma_1) = d(\text{Trans}_{\gamma_2} \circ \text{Trans}_{\gamma_1}, \text{Trans}_{\gamma_2 \circ \gamma_1}).$$

Birth and death mean failure to extend an object correspondence above a preregistered coverage threshold; split and merge mean a change in the cardinality of non-gauge equivalence classes under a fixed correspondence rule. They are empirical outcomes, not post-hoc explanatory escape hatches.

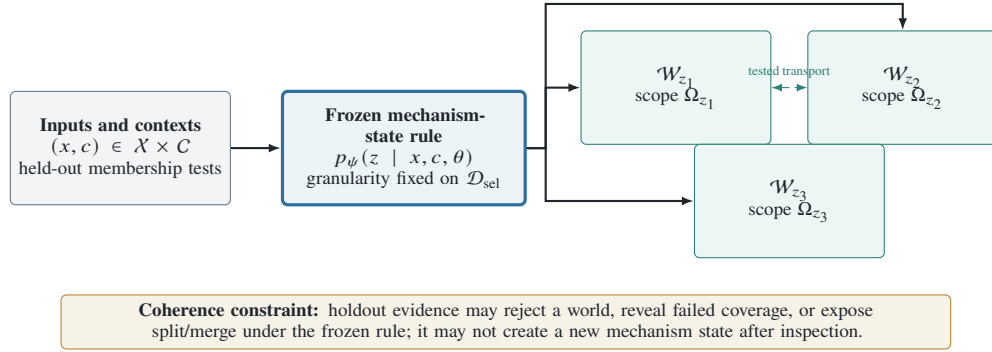


Figure 4: A coherent family of implementation worlds. Input- and context-dependent switching is permitted only through a preregistered or selection-frozen mechanism-state rule. Each state has a stable scope, and cross-context transport is a separately tested partial correspondence. This prevents “one post-hoc world per input” from making the theory vacuous.

3 Mechanistic inductive bias: the forward theory

Classical inductive bias is usually observed through functions learned, sample efficiency, or generalization. The present framework asks for the induced distribution over *implementations* among models that meet a declared behavioral criterion.

Definition 6 (Implementation equivalence level). Let Λ specify the claim type, gauge, scope, reader/intervention family, and tolerance used to compare implementation worlds. Write

$$\mathcal{W} \equiv_\Lambda \mathcal{W}'$$

when the two worlds satisfy the declared equivalence criterion. Examples include behavioral, abstract-algorithm, object/relation, tracker-family, physical-coordinate, and actionability-profile equivalence. These are distinct relations, not steps in a total order.

Definition 7 (Mechanistic inductive bias). Fix architecture syntax $\mathbf{S}_0$, training environment $\mathcal{E}$, comparison level Λ , and evaluation scope. Training and coherent mechanism selection induce

$$\xi \sim p_\Xi, \quad \theta \sim \text{Train}(\mathbf{S}_0, \mathcal{E}, \xi), \quad z \sim p(z \mid \theta, x, c),$$

and hence a probability law

$$\Pi_{\mathbf{S}_0, \mathcal{E}}^\Lambda := \mathcal{L}\left([\mathcal{W}_{\theta, z}^{\text{impl}}]_\Lambda\right)$$

over implementation-world equivalence classes. This law is the *mechanistic inductive bias* of the specified architecture–training system on the declared scope.

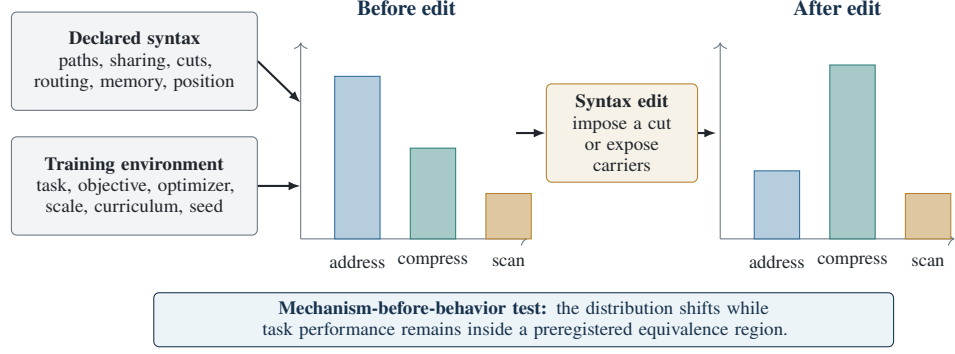


Figure 5: Mechanistic inductive bias is a distribution, not an architecture label. Bar heights are schematic probabilities over preregistered implementation classes, not reported data. A controlled syntax edit is informative when it predicts a directional redistribution, including before behavior separates.

The definition separates the target distribution from its measurement. Method syntax changes the evidence likelihood and recoverability of Π ; it does not become an argument of the generative law. The experimental frame and Λ are nevertheless necessary because there is no unscoped notion of implementation identity.

3.1 Three linked scientific problems

Forward prediction. Predict $\Pi_{S_0, \mathcal{E}}^\Lambda$ or a preregistered fingerprint

$$\phi(\mathcal{W}) = (\text{transport, cut use, routing, composition, failure interface, actionability})$$

from syntax and training before inspecting the final holdout internals.

Inverse recovery. Given a trained physical system and evidence produced under a method and experimental contract, recover a set or posterior over implementation classes:

$$\text{Recover} : (\mathfrak{P}_\theta, \mathfrak{Q}, \mathfrak{F}, \mathcal{D}) \longrightarrow \mathbb{P}([\mathcal{W}]_\Lambda \mid \mathcal{D}).$$

Architecture-edit causality. Under a coupled training design, change one syntax component and estimate the effect on mechanism fingerprints:

$$\Delta_{S_0 \rightarrow S'_0} \mathbb{E}[\phi(\mathcal{W})] = \mathbb{E}_{\Pi_{S'_0, \mathcal{E}}^\Lambda} [\phi] - \mathbb{E}_{\Pi_{S_0, \mathcal{E}}^\Lambda} [\phi].$$

When exact pairing is unavailable, the estimand is defined over matched training populations with explicit assumptions rather than called an individual-model causal effect.

Prediction 8 (Mechanism before behavior). For some operationally active syntax edits, the distribution of preregistered mechanism fingerprints will move detectably while task performance remains inside a declared equivalence region. The strongest version predicts the direction of that movement before holdout internal inspection. Failure to find such predictive surplus across matched tasks weakens the claim that architecture contributes anything beyond behavioral inductive bias.

3.2 Core predictive quantities

The framework has a clear empirical increment only if architecture information predicts mechanisms beyond task and training descriptors. Let $\mathcal{M}_0$ predict a fingerprint Y from $\mathcal{T}, \mathcal{E}$ and behavioral summaries, while $\mathcal{M}_1$ additionally receives S_0 . The primary architecture-aware advantage is the held-out proper-score difference

$$\Delta_{\text{arch}} = \mathbb{E}_{\text{hold}}[S(\mathcal{M}_1, Y) - S(\mathcal{M}_0, Y)].$$

Other population quantities are the entropy or dispersion of Π^Λ , the checkpoint at which a mechanism class becomes predictively stable, and the distribution of typed failure variables. These quantities require a declared candidate class or fingerprint; no universal scalar “mechanism distance” is assumed.

Theory claim. The first payoff is a separation, not a stereotype. For K independently queryable bindings, a fixed robust cut must distinguish q^K joint configurations, while K exposed carriers can store them separately and answer by selection. The architectures need not choose these implementations, but if they do, their dominant pressure moves from *compression* to *address selection*. [Theorem 16](#) states the result formally.

4 The inverse problem: measurement contracts and evidence licenses

4.1 Method syntax and independent data

Definition 9 (Method syntax). An interpretability method m has syntax

$$\mathfrak{Q}_m = (\text{Target}, \text{Obs}, \text{Do}, \text{Align}, \text{Agg}, \text{Scope}, \text{Cost}),$$

recording the claim it targets, observables, admissible interventions, alignment class, aggregation rule, scope, and supervision/search/compute budget.

Fitted methods use four disjoint roles:

$$\mathcal{D}_{\text{fit}} \rightarrow \text{fit probes/SAEs/alignments}, \quad \mathcal{D}_{\text{sel}} \rightarrow \text{select hyperparameters and candidate worlds},$$

$$\mathcal{D}_{\text{eval}} \rightarrow \text{estimate preregistered claims}, \quad \mathcal{D}_{\text{hold}} \rightarrow \text{test unseen interventions, compositions, and transports}.$$

Donor selection, prompt search, automated-agent trajectories, and stopping decisions belong to fitting or selection. Reusing them for final evidence requires selective-inference correction and is otherwise prohibited.

4.2 Intervention and identifiability contracts

Definition 10 (Intervention contract). A causal claim is indexed by

$$\mathfrak{C}_{\text{do}} = (\text{Estimand}, \text{Outcome}, \text{Treatment}, \text{Reference}, \text{Donor}, \text{Coupling}, \\ \text{Support}, \text{Reader}, \text{Interactions}, \text{Estimator}, \varepsilon, \delta, \text{Correction}, \text{Stopping}).$$

The contract states the causal estimand; outcome and native reader; treatment/reference states; donor-recipient coupling; admissible support and off-manifold diagnostics; interaction order; estimator; equivalence or superiority tolerance ε ; uncertainty δ ; multiple-testing correction; and stopping rule.

Definition 11 (Identifiability contract). $\mathfrak{C}_{\text{id}}$ specifies the candidate-world class, alignment family, allowed gauges, intervention and reader families inherited from $\mathfrak{C}_{\text{do}}$, complexity prior, scope, alternative hypotheses, and an observational/interventional indistinguishability threshold. It answers when two candidate worlds can be separated. It does not redefine a single causal effect.

The contracts compose as follows: $\mathfrak{C}_{\text{do}}$ defines the evidence likelihood for each candidate world; $\mathfrak{C}_{\text{id}}$ defines the equivalence relation and posterior/model-selection problem over candidates.

4.3 The measurement-to-ontology firewall

Firewall principle. An interpretability output is evidence licensed by a declared method syntax and experimental contract. It is not, by itself, a virtual object, native variable, causal mechanism, or privileged coordinate system of the model.

The firewall is not skepticism about measurement. It is a type discipline: a probe result, sparse code, attribution score, interchange effect, reconstruction, and transplant answer different questions. Promotion to a stronger claim requires the missing experiment rather than a change in vocabulary.

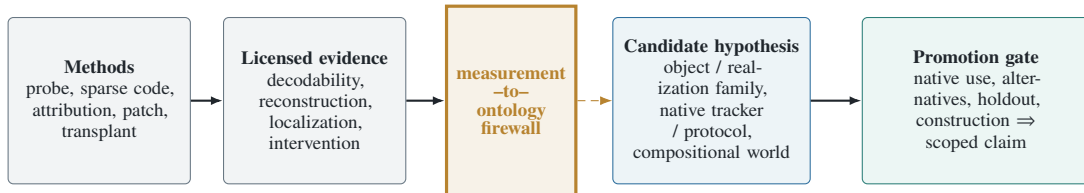


Figure 6: The measurement-to-ontology firewall. Methods generate licensed evidence, not ready-made ontology. Stronger implementation claims cross the firewall only through the additional native-use, alternative-model, structural-holdout, and constructive evidence appropriate to that claim.

Table 3: Default claim licenses. A method may exceed its default only by supplying the missing evidence.

Evidence	Default license	Does not by itself license
Held-out probe	Decodability in a specified class	Native use, causality, natural coordinates
Sparse dictionary	Sparse reconstruction in a chosen basis	Causal-variable, Jacobian, or pathway sparsity
Attribution/mask	A declared localization estimand	Necessity, sufficiency, or interaction completeness
Natural-donor interchange	Counterfactual use within support	Global identity or unique mechanism
Ablation/rescue	Scoped necessity/sufficiency	Portability or compositional closure
Reconstruction / transplant	Constructive actionability	Metaphysical uniqueness
Withheld composition	Typed compositional prediction	Higher context laws outside the test scope

Evidence profiles, not a scalar ladder. Interpretability targets are neither a total order nor a mathematically defined lattice. We report a profile

$$\mathbf{e} = (e_{\text{read}}, e_{\text{dec}}, e_{\text{loc}}, e_{\text{use}}, e_{\text{nec}}, e_{\text{suf}}, e_{\text{act}}, e_{\text{trans}}, e_{\text{comp}}, e_{\text{id}})$$

whose coordinates are separately defined estimands with uncertainty. A componentwise partial order is used only when two profiles share frame, scale, and tolerance. No “license gap” is formed by subtracting incomparable coordinates. Readability, actionability, reconstruction, and transport can be incomparable.

Four sparsities. The evidence synthesis motivates four distinct sparsities:

$$S_{\text{act}}, \quad S_{\text{var}}, \quad S_{\text{jac}}, \quad S_{\text{path}},$$

for activations, realized causal variables, Jacobian/computation, and physical pathways. Their units and failure modes differ:

S_{act} — **activation sparsity.**

Few coordinates are nonzero in a chosen activation basis. It licenses an economy of description in that basis, not a claim about causal variables.

S_{var} — **realized-variable sparsity.**

A small set of interventionally distinguished variables accounts for the tested counterfactuals. It depends on the intervention/readout frame and does not imply a sparse physical path.

S_{jac} — **computational sparsity.**

Few input-state directions exert locally non-negligible effects on declared downstream variables, for a stated Jacobian or finite-difference scale. It can be dense even when activations are sparse.

S_{path} — **pathway sparsity.**

A small physical coalition of edges, heads, neurons, or modules carries the tested causal effect under an interaction-aware estimand. Redundancy and self-repair can make a sparse native protocol look dense under isolated ablation.

None implies another without additional assumptions. Sparse autoencoders are therefore candidate coordinate systems and sensors, not privileged implementation worlds (Cunningham et al., 2023; Gao et al., 2024a; Farnik et al., 2025; Arora et al., 2026; Mueller et al., 2025; Wu et al., 2025b). The distinction is experimentally productive: a method should be evaluated against the sparsity it actually optimizes, not rewarded for a different sparsity by verbal transfer.

4.4 Multi-axis evidence coding

Single letters are replaced by

$$\mathbf{g}(s, k) = (I_{\text{int}}, I_{\text{do}}, I_{\text{match}}, I_{\text{gt}}, R, T, V),$$

for source s and claim k . The first four coordinates are binary: internal measurement, intervention, architecture-matched alternatives, and mechanism ground truth. $R \in \{0, 1, 2, 3\}$ records replication/task/model breadth; $T \in \{0, 1, 2, 3\}$ records theory-assumption strength, with higher values meaning stronger assumptions; $V \in \{\text{preprint, conference, journal, book}\}$ records publication status. Mixed labels such as “A/B” are disallowed. The supplement records each cited claim at this resolution.

5 Structural constraints

This section first records standard structural facts in the typed form needed by the research programme, then uses them to derive a carrier-cut separation for independently queryable bindings. Reachability is standard in structural causal and computation graphs; predictive quotients are related to automata minimization, predictive-state representations, and bisimulation; quotient descent is elementary (Pearl, 2009; Nerode, 1958; Singh et al., 2012; Ferns et al., 2004; Geiger et al., 2023). The new role of these ingredients is to connect memory organization to a directional difference in scaling pressure and failure interface.

5.1 Typed reachability

Lemma 12 (Communication-path necessity). *Let a deterministic typed acyclic computation graph have structural equations*

$$v = F_v(\text{pa}(v))$$

for every node v . Add explicit nodes or hyperedges for residual/skip paths, global pooling or normalization, cache state, cross-sample statistics, and shared randomness. If no directed path connects intervention target u to outcome node v , then any two admissible interventions on u with all other exogenous variables coupled identically produce the same v . In a stochastic graph, the corresponding interventional distributions are equal under the same coupling and structural assumptions.

Proof. Order nodes topologically. Nodes outside the descendants of u have identical parents and exogenous variables under both interventions, hence identical values by induction. Since v is not a descendant of u , it is invariant. The stochastic statement follows by conditioning on, or identically coupling, exogenous variables. $\square$

Corollary 13 (Local path lower bound). *If each layer permits edges of metric radius at most r_ℓ , an effect across distance d requires a path satisfying $\sum_{\ell=1}^L r_\ell \geq d$. This is only a path-existence lower bound. It does not imply that a detectable or usable effect grows smoothly or monotonically with d .*

5.2 Cut factorization and robust capacity

Let a history h cross a declared cut through state $c(h) \in C$, and let $\Sigma(h)$ denote its vector of future interventional outcome distributions under a specified experiment family. The predictive equivalence is

$$h \sim_\Sigma h' \iff \Sigma(h) = \Sigma(h').$$

Proposition 14 (Predictive cut factorization). *If $\Sigma(h)$ depends on h only through $c(h)$, then the quotient map $q_\Sigma : h \mapsto [h]_{\sim_\Sigma}$ factors through the cut image: there exists a unique map $\bar{q}$ on $c(\mathcal{H})$ such that*

$$q_\Sigma = \bar{q} \circ c.$$

The proposition does not assert that the learned cut state is minimal, disentangled, or sufficient for outcomes outside the declared experiment family.

Proof. If $c(h) = c(h')$, the assumption gives $\Sigma(h) = \Sigma(h')$, so $h \sim_\Sigma h'$. Therefore $\bar{q}(c(h)) = [h]_{\sim_\Sigma}$ is well-defined and unique on the cut image. $\square$

Finite width alone does not imply finite information capacity for real-valued states. A capacity claim requires noise, precision, robustness, or a mutual-information constraint.

Proposition 15 (Robust cut capacity). *Suppose q^K task configurations must be distinguished after a cut. Let their cut codes lie in a metric set (C_{reach}, d) and require pairwise margin greater than 2γ for a downstream reader to be robust to perturbations of radius γ . Then*

$$q^K \leq N_{\text{pack}}(C_{\text{reach}}, d, 2\gamma),$$

where N_{pack} is the packing number. If C_{reach} lies in a d -dimensional Euclidean ball of radius R , then

$$K \log q \leq d \log \left(1 + \frac{2R}{\gamma} \right).$$

Alternatively, if decoding error is at most α for a uniformly distributed configuration Y , Fano's inequality implies

$$(1 - \alpha)K \log q - h_2(\alpha) \leq I(Y; C).$$

Proof. Robustly distinguishable codes form a 2γ -packing. The Euclidean bound is the standard volumetric packing bound. The information statement is Fano’s inequality rearranged. $\square$

The empirical question is consequently not whether a vector has “finite width,” but which robust packing or information bound applies under declared precision, noise, margin, reader class, and error tolerance.

Theorem 16 (Carrier–cut separation for independently queryable bindings). *Let a task present K independently variable bindings $Y = (Y_1, \dots, Y_K) \in [q]^K$, followed after a memory boundary by a query $I \in [K]$ whose required answer is Y_I .*

1. *Suppose every influence of Y on the answer crosses a single state $C(Y) \in C$, and correctness must be robust to perturbations of radius γ at that cut. Then the q^K cut codes must form a 2γ -packing:*

$$q^K \leq N_{\text{pack}}(C_{\text{reach}}, d, 2\gamma).$$

In particular, any architecture family whose robust cut packing number is bounded independently of K must eventually incur nonzero worst-case error.

2. *Suppose instead that the memory boundary exposes an extensive carrier family $C_K = (C_1, \dots, C_K)$, each carrier has q robustly distinguishable states, and the query node has a permitted path to every carrier. Then an exact implementation exists: write Y_i into C_i , use I to select C_I , and emit its value. The required number of carriers grows linearly with K while the nominal query-to-carrier path length need not grow with K .*

Proof. For the first statement, if $Y \neq Y'$, then some coordinate i differs. Because query $I = i$ must return different answers, the two cut codes must be distinguishable by the downstream reader. Robustness to radius γ requires their perturbation balls to be disjoint, hence their distance exceeds 2γ . All q^K assignments therefore form a packing. For the second statement, choose q robust codes for each carrier, store coordinate Y_i in carrier i , and let the query select and decode carrier I . This construction uses K carriers and the declared direct path. $\square$

Corollary 17 (Failure-interface displacement). *For the task family in Theorem 16, increasing K places exponential distinguishability pressure on a fixed robust cut. An extensive addressable carrier family removes that particular fixed-cut lower bound by growing memory with K , but introduces a different empirical interface: selection among carriers and delivery to the downstream consumer. Thus an architecture edit can relocate the dominant failure pressure without changing low-load accuracy.*

Remark 18. The theorem is a structural separation, not a claim that every recurrent model uses a minimal compressed state or every Transformer learns source-local records. A recurrent architecture may enlarge or externalize its state; an attention architecture may compress history. The prediction concerns the resource scaling and failure interfaces of the two implementation classes.

5.3 Typed quotient descent and composition

Proposition 19 (Typed quotient descent). *Let $r : X \rightarrow Y$ and $s : Y \rightarrow Z$ be typed physical processes, with equivalence relations $\sim_X, \sim_Y, \sim_Z$. If*

$$x \sim_X x' \Rightarrow r(x) \sim_Y r(x'), \quad y \sim_Y y' \Rightarrow s(y) \sim_Z s(y'),$$

then unique maps

$$\bar{r} : X/\sim_X \rightarrow Y/\sim_Y, \quad \bar{s} : Y/\sim_Y \rightarrow Z/\sim_Z$$

are induced by $\bar{r}([x]) = [r(x)]$ and $\bar{s}([y]) = [s(y)]$, and

$$\overline{\bar{s} \circ \bar{r}} = \bar{s} \circ \bar{r}.$$

Proof. The preservation assumptions make both maps independent of representative. For every $[x]$,

$$\overline{\bar{s} \circ \bar{r}}([x]) = [s(r(x))] = \bar{s}([r(x)]) = (\bar{s} \circ \bar{r})([x]).$$

$\square$

Corollary 20 (Symmetry action on a quotient). *If a group acts equivariantly on typed states, interventions, and outcomes, future-intervention equivalence is preserved and the group acts on the quotient by $g \cdot [x] = [g \cdot x]$.*

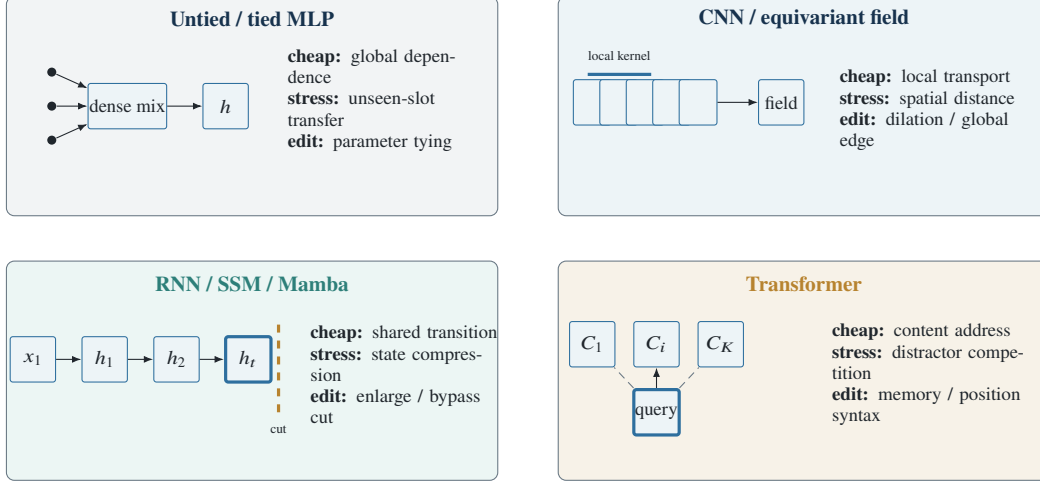


Figure 7: Four core causal syntaxes and their mechanism priors. The cartoons do not assign a unique ontology to an architecture. They show where each syntax makes a candidate organization relatively direct, where scaling pressure is expected to enter, and which controlled edit can test the account.

Remark 21. [Corollary 20](#) supplies an action on quotient classes. It does not identify orbit members as one object, guarantee low physical transport defect, or guarantee that a chosen method recovers the action. Those are separate empirical claims.

For approximate symmetry, let each transition have equivariance defect at most η_ℓ and Lipschitz constant L_ℓ . A length- n composition obeys the telescoping bound

$$\eta_{1:n} \leq \sum_{k=1}^n \eta_k \prod_{j=k+1}^n L_j.$$

Approximate identity is represented by a pseudometric or cover, not by an ε -threshold relation falsely assumed transitive.

6 Architecture-conditioned deductions

The structural results eliminate impossible mechanisms; they do not select a unique remaining world. An architecture-conditioned deduction must therefore be stated as a hard consequence, an evidence-supported tendency, or an open conjecture. The probabilistic target is

$$\Pi_{S_0, \varepsilon}^\Lambda,$$

with scope, competing explanations, and a syntax edit that can move the distribution. This section focuses on four core contrasts. Evidence cited here is claim-scoped; evidence axes and publication status appear in the integrated appendices and source files.

6.1 Untied and tied MLP baselines

Syntax. An untied dense MLP has short nominal path length and fixed potential support, but no canonical coordinate transport. A tied or equivariant MLP adds sharing or a group action. A static token mixer has a fixed mixing matrix, yet nonlinear gates can still produce strongly input-dependent *effect magnitudes*; fixed support must not be equated with low causal adaptivity ([Ravanbakhsh, 2020](#); [Wang et al., 2022](#); [Yu et al., 2021](#)).

Conditional prior. Cheap global dependence and costly exchangeability are the baseline expectation. The same abstract operation at two coordinates need not share a tracker unless tying, data symmetry, or optimization creates one.

Prediction 22 (MLP transport prior). On an exchangeable-slot task, an untied MLP should have a larger held-out slot-transport defect than a parameter-tied control at matched task performance. The claim concerns transport, not a unique “holistic ontology.” It is rejected if the difference lies within a preregistered equivalence margin.

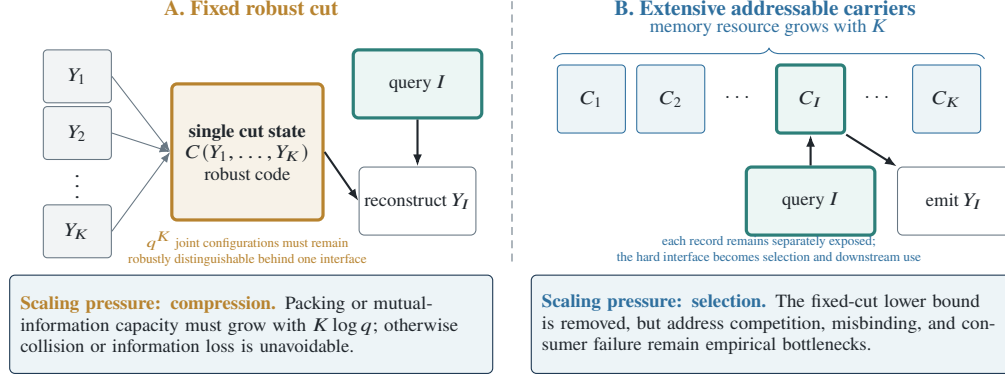


Figure 8: The carrier–cut separation. The same retrieval behavior can be implemented through two memory organizations with different resource scaling and different failure interfaces. [Theorem 16](#) makes the left-hand pressure formal and gives the right-hand construction. It does not assume which organization training selects.

Decisive edit. Introduce tying while keeping potential reach fixed. A reduction in cross-slot tracker defect before, or without, an accuracy change supports the syntax account. Dynamic mixing is separate: it tests conditional source selection, not transport by itself.

6.2 Convolutional networks

Syntax. CNNs combine local communication and spatial parameter sharing. Exact translation equivariance depends on boundary, stride, pooling, sampling, and readout; zero padding can leak absolute position ([Cohen and Welling, 2016](#); [Islam et al., 2021](#); [Luo et al., 2016](#)). Group-equivariant CNNs make a larger exact action explicit, while residual and U-Net paths alter reach and cuts.

Conditional prior. When task, data, boundary, intervention, and reader preserve a group action, position-indexed causal fields and reusable local trackers should be cheaper than coordinate-specific rediscovery. [Corollary 20](#) only guarantees an action on a quotient; treating orbit-related realizers as the same object additionally requires native-reader interchange and a declared identity criterion.

Prediction 23 (CNN transport and propagation). In successful models on a task that requires a relation across controlled distance, symmetry-preserving CNNs should show: (i) lower shift-transport defect than untied locally connected controls; and (ii) a positive relationship between required distance and the earliest *detectable native causal effect*, subject to censoring and effective receptive field. Neither statement follows from path existence alone.

Boundaries. Shape/texture preference is not a syntax invariant: data and augmentation can reverse or suppress it ([Geirhos et al., 2019](#); [Hermann et al., 2020](#); [Burgert et al., 2025](#)). Hierarchical composition is a prior, not a theorem; pooling, shortcuts, and global readouts can bypass it.

Decisive edits. Compare circular, valid, and zero-padded variants; add dilation or a global edge while preserving the task; and compare tied with locally connected untied filters. The mechanism prediction is directional: padding should increase coordinate information; dilation/global edges should reduce causal mediation depth; tying should reduce transport defect.

6.3 Recurrent and state-space models

Shared cut. RNNs and SSMs route all prefix influence through a recurrent state at every time. [Proposition 14](#) makes this a factorization statement, not a claim that the state is minimal or predictive. Time-shared parameters do not guarantee time-invariant trackers: gates, boundary state, input distribution, and saturation change effective maps ([Hochreiter and Schmidhuber, 1997](#); [Downey et al., 2017](#); [Venkatraman et al., 2017](#)).

RNN prior. Task-sufficient cut states and shared transition protocols are relatively cheap. Under increasing concurrent binding load, robust finite-resource models should exhibit mergers, overwrite, or forgetting when the required distinguishability exceeds the cut’s packing/information budget. Fixed dimension alone is not the premise.

Linear SSM prior. For

$$h_t = Ah_{t-1} + Bx_t, \quad y_t = Ch_t + Dx_t,$$

past inputs enter as mode-filtered superpositions. Retention lifetimes and terminal reconstruction are therefore natural candidates, but their realization depends on spectrum, discretization, precision, parameterization, and optimization (Gu et al., 2021; Wang and Li, 2024; Jelassi et al., 2024; Merrill et al., 2024).

Selective SSM precision. Canonical Mamba keeps a learned continuous-time A and skip D while making the discretization step Δ_t and input/output projections B_t, C_t input-dependent:

$$\bar{A}_t = \exp(\Delta_t A), \quad h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = C_t h_t + D x_t.$$

A short convolution precedes the selective scan (Gu and Dao, 2023). Selectivity changes retention and readout but does not create unbounded robust addressable memory.

Prediction 24 (Cut versus address). On matched associative-recall tasks, recurrent models should more often place decisive information in the evolving or terminal recurrent state, while token-indexed Transformers should more often preserve source-local records. This is a scoped distributional prediction, not an SSM ontology. Evidence from AR/ATR supports it only for the tested architectures and settings (Arora et al., 2025).

Transformer and Mamba may nonetheless implement analogous induction algorithms while differing in temporal deployment; SAE similarity alone does not establish causal identity (Wang et al., 2024). State size, short convolution, selectivity, and initialization are causal edits that should move the mechanism distribution (Trockman et al., 2024).

6.4 Transformers

Syntax. Transformers provide token-indexed residual carriers, short nominal path length under full attention, content-dependent QK selection, and OV-dependent written content (Vaswani et al., 2017; Elhage et al., 2021). Attention weights specify neither causal edge effects nor downstream use. The empirically effective graph depends on values, output projection, competing heads, residual paths, and native readers. KV cache is an inference implementation of growing token memory, distinct from the abstract full-sequence training graph.

Relative prior. Content-addressed protocols are cheap, not compulsory. A useful candidate unit is typed:

$$\text{writer} \rightarrow \text{carrier} \rightarrow \text{address/query} \rightarrow \text{transform/write} \rightarrow \text{consumer}.$$

Induction circuits and variable-binding studies provide concrete multi-component instances (Olsson et al., 2022; Wu et al., 2025a; Merullo et al., 2024). The same architecture can also learn scans, compressed states, convolution-like maps, heuristics, or input-conditioned mixtures (Li et al., 2025; Cordonnier et al., 2019; Li et al., 2021; Pîslar et al., 2025).

6.4.1 Worked example: symbolic dereferencing

Consider a controlled program

$$[A := v_1] [B := v_2] \cdots [Q := B] [\text{answer} = ?],$$

with optional chains such as $Q := C$ and $C := B$. Surface names, values, positions, and irrelevant assignments are independently randomized. The *task specification* contains bindings and typed dereference operations. It does not specify how a trained Transformer implements them.

Competing implementation worlds. At least three candidates must be retained during discovery.

1. **Address protocol** $\mathcal{H}_{\text{addr}}$. Assignment sites preserve source-local records; later queries construct a role/address; QK-like selection locates a record; an OV-like map writes a value into a publication state; a downstream consumer uses it.
2. **Compressed-state or scan protocol** $\mathcal{H}_{\text{scan}}$. Assignments are successively folded into a task-sufficient state. The query is answered from that state rather than by source-local dereference.
3. **Heuristic or mixture** $\mathcal{H}_{\text{mix}}$. Absolute position, recency, name frequency, or a small candidate set supports a shortcut on part of the distribution, with a different protocol activated elsewhere.

The candidate class also includes refinements and observationally equivalent variants. Its granularity and coding language are frozen on $\mathcal{D}_{\text{sel}}$; otherwise every error can be redescribed as a new world.

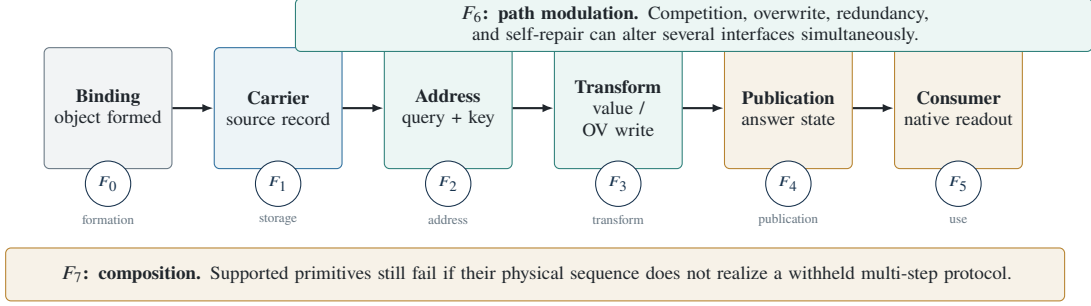


Figure 9: A stage-resolved Transformer protocol map. Information can be present at one interface and fail at a later one; competition and repair can alter several stages simultaneously. The labels are causal hypotheses to be distinguished by interventions, not mutually exclusive post-hoc error names.

Table 4: Protocol audit for the Transformer dereferencing example. Rows are candidate causal interfaces, not mutually exclusive labels or an assumed temporal hierarchy.

Interface	Candidate evidence	Discriminating intervention	What a positive result licenses
Formation	Abstract value or binding can be decoded at a nominated site	Natural-donor interchange across matched assignments	Scoped realization at the tested site; not downstream use
Carrier/storage	Information persists under pulse-chase and matched perturbations	Replace or erase the nominated record while holding query and reader fixed	Storage dependence within the declared support
Address/query	Query and key variables predict source selection	Swap address variables while preserving values and positions	Address-mediated selection, conditional on downstream write
Transform/write	Selected source affects the proposed publication state	Intervene on value/OV map with source selection held fixed	Typed source-to-publication transformation
Publication	A recoverable answer state is written before final readout	Patch publication state while leaving earlier records unchanged	Availability to a declared downstream consumer
Consumer/readout	Native reader uses the publication state	Disable and rescue reader with matched publication donors	Scoped consumption/actionability
Competition/repair	Effects change with candidate similarity, count, or co-ablation	Factorial distractors; Harsanyi/ANOVA coalitions; conditional co-ablation	Interaction, redundancy, or self-repair in the tested coalition
Composition	Primitive dereferences are individually supported	Withhold and test typed two-step dereference $g \circ f$	Compositional prediction, not a universal algebra

Separation of discovery and test. Probes, sparse coordinates, attention patterns, and attribution are used on $\mathcal{D}_{\text{fit}}$ only to nominate carrier, address, write, and consumer candidates. Hyperparameters, candidate partitions, and interaction coalitions are selected on $\mathcal{D}_{\text{sel}}$. Native-donor interventions and stage-specific rescues estimate declared effects on $\mathcal{D}_{\text{eval}}$. The decisive tests use $\mathcal{D}_{\text{hold}}$: unseen names and values, unseen absolute positions, longer contexts, factorially controlled distractor count and similarity, and withheld dereference compositions. No representation discovered on the holdout may enter the claim.

Evidence calibration. Induction circuits and controlled variable-binding systems support the possibility of $\mathcal{H}_{\text{addr}}$, while state-tracking experiments show that a Transformer can select different algorithms under curriculum, initialization, or task variation (Olsson et al., 2022; Wu et al., 2025a; Li et al., 2025; Zhong et al., 2023). Long-context studies establish position and utilization failures, but do not by themselves identify which interface failed (Liu et al., 2024; Gao et al., 2024b; Du et al., 2025). Distributed intervention and self-repair results motivate coalition tests rather than singleton rankings (Cheng and Zhang, 2026; Gong et al., 2026; Vaidyanathan et al., 2026). The worked example therefore demonstrates the framework; it does not establish that natural-language Transformers universally use an address ontology.

Retrieval geometry. Full attention removes a distance-growing *reachability lower bound*; it does not remove position, length, normalization, competition, or reader effects. The testable estimand is not an unspecified function

$P(\text{error}) = f(\cdot)$, but a preregistered factorial contrast over absolute position, relative distance, context length, candidate count, candidate similarity, and position syntax. Long-context behavior depends on absolute position and length even under successful retrieval (Liu et al., 2024; Du et al., 2025; Gao et al., 2024b).

Prediction 25 (Reach versus retrieval). After matching absolute position, context length, and distractor statistics, increasing relative source distance should not produce the path-length scaling required by a local CNN. Increasing candidate similarity or count should increase address competition. Superiority or equivalence is assessed with preregistered contrasts, not qualitative “low/high” labels.

Storage/use dissociation. Decodable target information can coexist with incorrect output (Gao et al., 2024b; Karim et al., 2026; Basu et al., 2026). This licenses a dissociation, not the conclusion that editing a query or reader will repair the model. Repair requires stage-specific intervention.

Protocol interactions. Single-component patching can mis-rank conjunctive, redundant, or self-repairing systems (Vaidyanathan et al., 2026; Cheng and Zhang, 2026; Gong et al., 2026). Group intervention is expected to dominate isolated intervention only under a declared interaction structure. We use Harsanyi dividends,

$$\Delta_H(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} \Delta(T),$$

or a preregistered causal interaction estimand, rather than the uncalibrated difference between a coalition and the sum of singleton effects.

Position syntax. Object transport depends on the joint position syntax

$$\text{Pos} = (\text{encoding, mask, normalization, BOS/sinks, residuals, training range}).$$

Absolute, relative, rotary, randomized, and no-explicit-encoding systems have different behavioral length-generalization profiles (Haviv et al., 2022; Kazemnejad et al., 2023; Ruoss et al., 2023; Sinha et al., 2022; Ruscio et al., 2026). These results do not by themselves establish causal object identity across positions. That requires a natural-donor transport test with the same abstract counterfactual at source and destination.

World distributions. Mechanism diversity must be reported at a preregistered conservation level: behavior, abstract algorithm, object structure, tracker family, or coordinate. Entropy is meaningful only after fixing this equivalence and reporting sensitivity to alternate partitions. Curriculum, initialization, checkpoint, scale, and input can change the mechanism (Zhong et al., 2023; Li et al., 2025; Yin and Steinhardt, 2025; Píslar et al., 2025).

Interface-typed failures. We use a multi-label causal graph, not a mutually exclusive “earliest failed interface.” Candidate failure variables are formation, carrier/storage, address/query, transform/write, publication, consumer/readout, competition/overwrite, and composition. Several may co-occur or be masked by repair. A posterior over them is reported only if a generative model and likelihood are specified; otherwise the output is a vector of intervention results.

Prediction 26 (Transformer protocol prior). At matched performance, Transformer errors on binding/retrieval tasks should be more recoverable by address-, competition-, publication-, or consumer-targeted interventions than recurrent errors whose relevant information has already collapsed before query time. The comparison is distributional and task-scoped; formation and storage failures remain possible in Transformers.

Decisive syntax edits. Compare full with local attention at fixed position syntax; change absolute, relative, rotary, randomized, and no-explicit-position systems; compress or evict token memory; and add a recurrent bottleneck while retaining comparable parameterized transforms. The predicted quantities are not merely accuracy and attention shape, but source-local storage mass, tracker transport, competition effects, protocol composition, and the distribution of interface-typed failures.

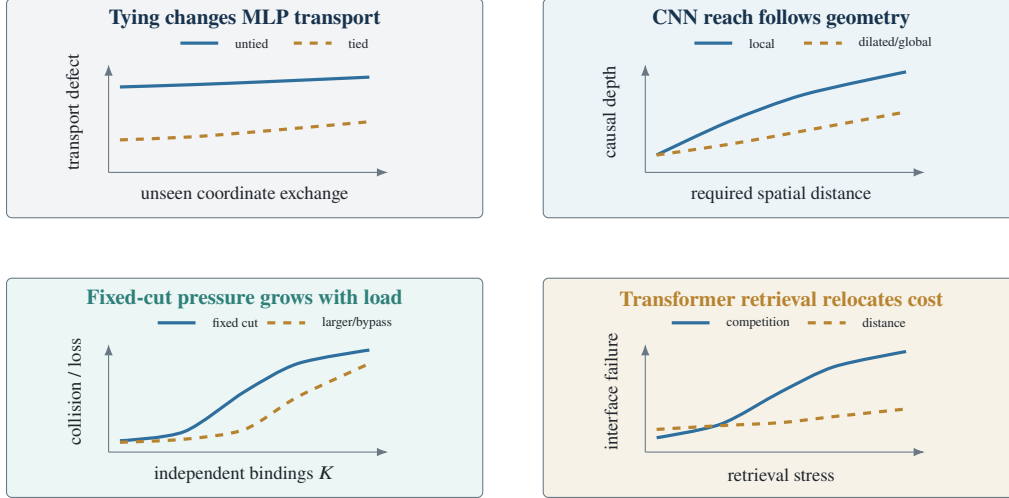


Figure 10: Predicted qualitative signatures, not reported data. Each panel pairs a stress variable with a syntax edit that should move the curve: tying should improve MLP transport; dilation or global edges should reduce CNN mediation depth; additional or bypassed state should delay fixed-cut collision; and Transformer retrieval should be more sensitive to candidate competition than to controlled source distance. The decisive evidence is a directional shift, not the architecture label.

How to read the atlas. The four panels are not phenomenological portraits of architecture classes. They are *response-shape predictions*. A syntax claim earns explanatory credit only when an intervention that changes the named structural resource changes the corresponding curve in the preregistered direction. Thus the scientific object is not a static label such as “CNN-like” or “Transformer-like,” but a paired contrast

(stress response before edit, stress response after edit).

This packaging separates a merely compatible story from a causal theory of mechanistic inductive bias.

Why the signatures form one theory. The panels share a common grammar: identify the carrier of task-relevant distinctions, locate the interface through which those distinctions must pass, increase a stress variable, and edit the syntax that prices the candidate implementation. Tying changes the cost of transport; dilation changes the cost of reach; state expansion changes the cost of compression; and attention or memory edits change the cost of selection. The curves differ because the same abstract burden is paid at different physical interfaces.

6.5 Extension syntaxes in the same language

The core contrast is not limited to four architecture families. Three extensions expose different causal geometries while preserving the same question: what is cheap to carry, where is the mandatory interface, what failure should scale first, and which syntax edit should move it? Detailed technical boundaries and evidence appear in [Appendix C](#); the mechanistic predictions belong to the main theory.

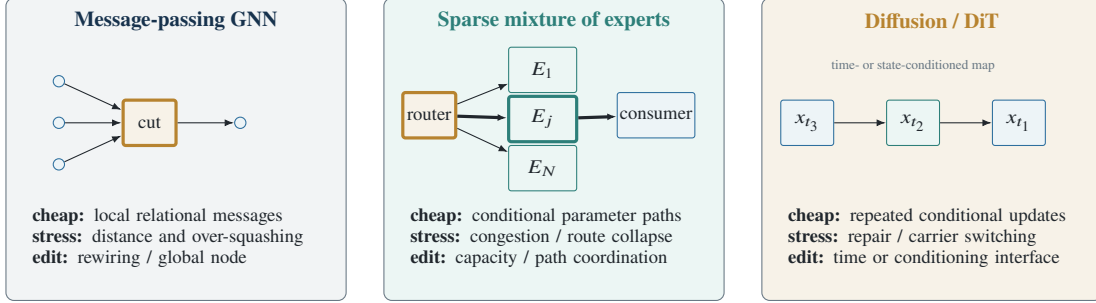


Figure 11: The extension panorama. GNNs, sparse MoE systems, and diffusion/DiT models instantiate graph-local aggregation, discrete conditional paths, and repeated state- or time-conditioned updates. The panels state conditional mechanism priors and decisive edits, not established architecture ontologies.

Graph-local aggregation. For a message-passing GNN, the native carrier is indexed by nodes, edges, and neighborhoods, while graph distance bounds the earliest possible mediation. The distinctive pressure is not a generic “long-range problem”: information can be lost through over-squashing at a graph cut, homogenized through oversmoothing, or remain unidentified because the message-passing class cannot separate the relevant structures. Rewiring, a global node, or an enlarged cut must therefore move the corresponding causal signature before it earns credit as an explanation.

Conditionally routed parameter paths. For sparse mixtures of experts, expert identity is not the computational object. The candidate mechanism is the complete router–expert–path–consumer protocol, including capacity constraints and token dropping. A semantic histogram over experts is weak evidence unless the route predicts held-out transformation and downstream use. Decisive edits change router coordination, active capacity, or expert paths while measuring congestion, path stability, collateral effects, and function-preserving transport.

Repeated time-conditioned worlds. For diffusion and DiT systems, a shared parameter block can implement different effective maps at different noise or flow times. The object of explanation is therefore a family $\mathcal{W}_{\ell,t,x,c}$, not a static layer atlas. Carrier switching, repair, amplification, and trajectory bifurcation are distinct hypotheses. Direct transport from t_1 to t_3 must be compared with transport composed through t_2 ; edits to time conditioning, backbone connectivity, or the conditioning interface should move the predicted trajectory-level failure.

7 A calibrated measurement language

A compact formula is not an operational metric until it specifies the estimand, estimator, null, tolerance, and power target. Table 5 gives the core cards. All estimators are cross-fitted: candidate discovery uses $\mathcal{D}_{\text{fit}}$ and $\mathcal{D}_{\text{sel}}$, while estimates use $\mathcal{D}_{\text{eval}}$ and structural generalization uses $\mathcal{D}_{\text{hold}}$.

Table 5: Core estimand cards. “Natural” means sampled from the preregistered support rather than optimized post hoc.

Target	Estimand	Estimator and null	Calibration
Symmetry transport	Difference in native outcome under transformed natural donor versus matched source counterfactual, minimized over declared gauge	Cross-fitted paired effect; null is equality to coordinate-shuffled or untied control	Equivalence margin ε_{sym} ; group sampling law; familywise correction; target power $1 - \beta$
Propagation	Slope/finite differences of first layer with native effect above ε_{eff} as task distance changes	Interval-censored regression because “not detected” is censored; null slope set in advance	Distance grid, effect threshold, power at each layer, and multiplicity correction
Routing	Four-vector: support turnover, normalized edge-effect drift, written-content drift, downstream-use drift	Natural-donor interventions; compare inputs matched on scale and geometry; attention weights are not effects	Separate equivalence margins; report vector, never a scalar routing score
Cut sufficiency	Expected proper-score regret for predicting future intervention outcomes from cut state in a fixed hypothesis class	Cross-fitted probabilistic decoder versus history-aware and label-shuffled controls	Decoder class, future-experiment law, train/eval split, confidence interval, and complexity penalty
Robust capacity	Packing lower bound or mutual information needed for K simultaneous variables	Adversarial/noise-radius test and Fano-compatible decoder bound	Precision, noise, margin, allowed error, reader class, and state scaling
Tracker transport	Change in typed counterfactual effect under a partial carrier correspondence	Span/correspondence matching; null is donor shuffle or gauge-only alignment	Coverage threshold, non-invertible cases, equivalence margin, and uncovered mass
Composition	Native-outcome loss of physical $r_g \circ r_f$ on withheld task composite $g \circ f$	Decode/realize through native outcome map $\omega_Z : Z \rightarrow O$; compare direct composite, shuffled order, and task-only baseline	Withheld composite family, type checks, equivalence/superiority margin, and sample size
Multiplicity	Lower bound on discovered non-gauge realization classes within a bounded search family	Bayesian or MDL model comparison after interchange-equivalence testing	Search class, gauge group, posterior threshold, and sensitivity analysis
Mechanism distribution	Predictive distribution over preregistered mechanism classes or an MDL partition	Hierarchical model across seeds/checkpoints/inputs; task-only versus syntax-conditioned log predictive density	Fixed equivalence level, held-out architectures/tasks, partition sensitivity, uncertainty

7.1 Typed composition

The physical and abstract outputs cannot be compared in one untyped loss. For typed trackers

$$r_f : X \rightarrow Y, \quad r_g : Y \rightarrow Z,$$

let $\omega_Z : Z \rightarrow O_Z$ be a native outcome map and let $v_{g \circ f}(u) \in O_Z$ be the task-specified counterfactual outcome. The held-out composition defect is

$$\Delta_o = \mathbb{E}_{(a,u) \sim p_{\text{hold}}} \ell_{O_Z}(\omega_Z(r_g(r_f(a))), v_{g \circ f}(u)).$$

The composite $g \circ f$, donor family, and metric are withheld from recovery. A low primitive error with high Δ_o is evidence for a parts atlas without compositional closure.

7.2 Interaction and self-repair

Coalition effects are not reduced to singleton rankings. For component set N , estimate the set function $\Delta(S)$ for preregistered coalitions and decompose it with Harsanyi dividends or a specified causal ANOVA. Positive high-order terms indicate conjunctive dependence; negative terms can indicate redundancy or repair. Conditional co-ablation tests components in the background where they are predicted to become necessary. Order is treated explicitly for writer-reader protocols.

7.3 No premature scalarization

Concurrent-variable stress produces a vector:

$$\mathbf{I}(K) = (\Delta_{\text{object}}(K), \Delta_{\text{tracker}}(K), \Delta_o(K), \text{FDR}_{\text{object}}(K), \text{coverage}(K)).$$

Arbitrary weighted sums are not used. Likewise, actionability is a vector of correction rate, collateral damage, robustness, transfer, and compute, rather than a subtraction from a localization score.

8 Evidence, counterevidence, and synthesis principles

The literature does not yet constitute a unified theory of architecture-conditioned internal computation. It provides fragments: matched architecture comparisons, mechanism case studies, formal constraints, representation analyses, intervention failures, and architecture edits. This section asks which parts of AVC are already well supported, which are scoped, and which remain conjectural. It is a selective synthesis of cited primary and formal sources, frozen July 27, 2026, not an exhaustive or PRISMA-complete review. The integrated audit files list every retained source, the exact claim for which it is used, evidence axes, publication status, and limitations.

8.1 Replicated or mature findings

1. **Behavior underdetermines representation and mechanism.** Probe controls, representation non-identifiability, and causal-abstraction work show that predictive fit alone does not establish native causal variables (Hewitt and Liang, 2019; Ravichander et al., 2021; Locatello et al., 2019; Geiger et al., 2023; Méloux et al., 2025; Sutter et al., 2025).
2. **Architecture components change structural affordances.** Locality/sharing, group equivariance, recurrent cuts, attention selection, and SSM state dynamics have clear formal or architectural consequences (Cohen and Welling, 2016; Ravanbakhsh, 2020; Vaswani et al., 2017; Gu et al., 2021; Gu and Dao, 2023).
3. **Position and context length affect Transformer retrieval.** Multiple studies show position-dependent long-context performance and position-system-dependent extrapolation (Liu et al., 2024; Haviv et al., 2022; Sinha et al., 2022; Kazemnejad et al., 2023; Ruoss et al., 2023; Du et al., 2025).
4. **Sparse reconstruction is not a privileged ontology.** Probe/benchmark comparisons, random-network results, neuron-basis circuits, and Jacobian sparsification separate activation sparsity from causal and computational sparsity (Heap et al., 2025; Arora et al., 2026; Farnik et al., 2025; Mueller et al., 2025; Wu et al., 2025b).

8.2 Strong but scoped causal findings

1. Matched AR/ATR experiments report induction-style source-local storage in tested Transformers/Based models and terminal-state computation in most tested SSMs; the claim is scoped to those tasks and models (Arora et al., 2025).
2. Transformer variable binding can use token-indexed addressable memory and specialized routing heads in controlled symbolic programs (Wu et al., 2025a).

3. Curriculum can shift a Transformer between state-tracking algorithms, and initialization/hyperparameters can produce different algorithmic solutions (Li et al., 2025; Zhong et al., 2023).
4. Natural-donor and multi-position interventions show that decodability can dissociate from use and that causal units can be distributed (Gao et al., 2024b; Cheng and Zhang, 2026).

8.3 Methodological cautions

1. Saliency, attention, probes, trained lenses, stitching, and sparse dictionaries estimate different targets and can pass misleading proxy tests (Adebayo et al., 2018; Jain and Wallace, 2019; Wiegrefe and Pinter, 2019; Belrose et al., 2023; Csiszarik et al., 2021; Chanin, 2026).
2. Single-mediator patching can contain interaction terms, and self-repair can hide backup pathways (Vaidyanathan et al., 2026; Gong et al., 2026).
3. Formal identifiability requires assumptions and does not ensure finite-sample or optimization recovery (Khemakhem et al., 2020; Schölkopf et al., 2021; Gamella et al., 2025).
4. Near-perfect decodability need not yield a reliable low-collateral causal handle (Basu et al., 2026).

8.4 Proposed synthesis hypotheses

The combined record supports several claims at different epistemic levels. They are not pooled into “established laws.” Table 6 separates structural consequences, methodological rules, scoped hypotheses, and acceptance criteria.

Table 6: Status-coded synthesis. “Record” summarizes why the proposition is worth testing; it is not a meta-analytic effect estimate.

Proposition	Current status	Supporting record	Decisive failure or test
Causal-geometry constraint	Structural consequence under declared equations	Reachability, exact symmetry, cuts, and typed composition follow from specified graphs and quotients	A claimed hard constraint disappears after all omitted paths, shared state, readers, and granularity are modeled
Measurement-to-ontology firewall	Methodological type rule	Probes, attribution, sparse codes, patching, and reconstruction license different claims and have distinct failure modes	A method-specific output predicts native use, transport, and composition without the supposedly missing evidence
Mechanistic-inductive-bias hypothesis	Central empirical hypothesis	Matched AR/ATR and state-tracking studies show architecture- and training-conditioned mechanism differences	A task-, resource-, behavior-, and training-only model matches syntax-conditioned held-out mechanism prediction within the equivalence margin
Syntax-edit response	Central causal test	Sharing, boundary, state, position, routing, and convolution components have known structural or scoped mechanistic effects	Operationally used edits fail to move the preregistered mechanism fingerprint
Cut-pressure hypothesis	Scoped architecture hypothesis	Recurrent cuts, graph bottlenecks, and finite robust packing create distinguishability pressure under load	Increasing cut capacity or bypassing the cut does not move the predicted failure profile
Failure-interface hypothesis	Open cross-architecture hypothesis	Retrieval, compression, propagation, routing, and trajectory failures occur in architecture-linked case studies	Task and load predict failure location as well as syntax across matched systems
Routing-nonsemanticity rule	Methodological caution	Attention, gates, and expert routes identify selection structure, not a complete transformation or consumer	Router identity alone predicts held-out function as well as the full route-transform-consumer protocol
Conservation-gradient hypothesis	Cross-model hypothesis	Abstract algorithms sometimes recur while timing, carriers, circuits, and coordinates differ	Tracker families or coordinates transfer at least as broadly as abstract algorithms under preregistered equivalence classes
Dynamic-world hypothesis	Scoped extension	Recurrent state, conditional routing, and diffusion time can change the active map despite shared parameters	A single static world predicts cross-regime interventions and transport within tolerance
Constructive closure	Acceptance criterion	Readability and localization can dissociate from low-collateral control; composition and transplant offer stronger tests	Readability/localization-selected accounts reconstruct and transport as reliably as accounts selected on structural holdouts

9 Decisive tests of the theory

The empirical protocol is an instrument for testing the theory, not the paper’s identity. Two results would provide the clearest initial evidence. First, a syntax-aware model should predict a mechanism fingerprint on held-out training populations better than a model given task, optimization, resource, and behavioral descriptors alone. Second, an operationally active syntax edit should move the fingerprint and its failure profile in a preregistered direction, including settings where accuracy remains equivalent. Appendix A gives the complete tasks, architecture contrasts, ground-truth regimes, data roles, estimators, power targets, and preregistration requirements.

Phase I combines three tasks—orbit transport, pointer lookup, and held-out typed composition—with four core architecture contrasts. It uses three evidential regimes: planted-mechanism organisms, trained synthetic models whose task is known but mechanism is not, and natural models with predictive and constructive rather than ontological scoring. Candidate mechanisms are nominated without holdout access and tested on unseen interventions, compositions, transports, and architecture edits.

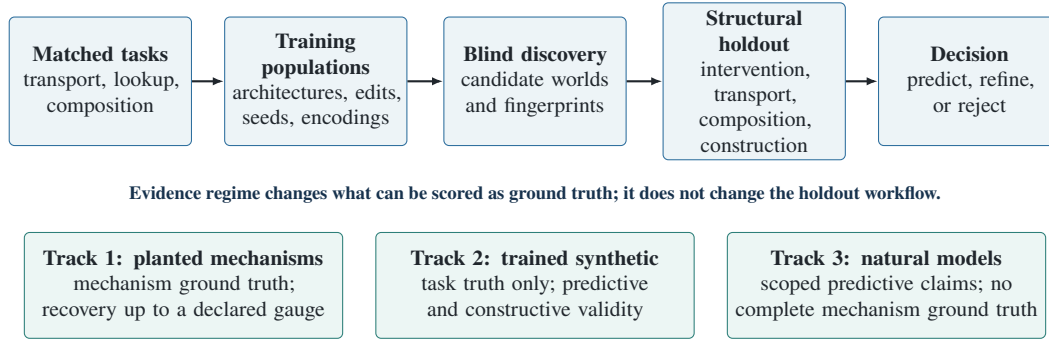


Figure 12: Decisive tests of the theory. Ground truth changes across tracks, but the logic does not: discover without holdout access, test structural predictions, and ask whether syntax and syntax edits add predictive value beyond task, training, behavior, and resource baselines.

The primary population-level question is the architecture-aware predictive advantage Δ_{arch} from section 3. The strongest causal question is whether a preregistered syntax edit moves $\phi(W)$ in the predicted direction while operational use of the edit is verified. The strongest inverse criterion is constructive closure: a recovered account predicts held-out intervention and supports selective rescue, composition, transport, or reconstruction under shift. Accuracy alone is never the primary endpoint.

10 Falsification and contradiction handling

A conditional prior is not decisively refuted by one failed task unless its scope included that task. The program is weakened when preregistered syntax variables fail to improve held-out prediction repeatedly across scoped tasks, or when syntax edits do not move the predicted mechanism fingerprint despite being operationally used.

Table 7: Predeclared diagnosis categories. They constrain, rather than excuse, theory revision.

Category	Evidence required	Consequence
Assumption failure	Demonstrate a missing edge, broken symmetry, inadequate precision/noise premise, or unscoped reader	Narrow theorem application; do not change the empirical target
Syntax effect absent	Edit is operationally active but fingerprint stays within null/equivalence region	Lower or reject the architecture prior
Training override	Same syntax, controlled training edit changes world distribution	Add the training variable; reduce architecture-only claim
Method mismatch	Independent methods targeting different estimands disagree as predicted by their contracts	Keep separate licenses; do not average scores
Non-identifiability	Candidate worlds make equivalent held-out predictions under the full contract	Return both in $\widehat{\mathfrak{M}}_{\varepsilon, \delta}$
Post-hoc rescue	New world class or threshold introduced after holdout inspection	Exploratory only; requires a fresh holdout

The strongest failure condition is:

Across preregistered tasks and architecture edits, a task-and-training-only baseline predicts mechanism fingerprints, held-out interventions, and failure locations as well as a syntax-conditioned model within the equivalence margin.

That result would reject the program’s central predictive increment even if architecture remains useful for performance.

11 Discussion

11.1 What the framework does and does not claim

AVC changes the dependent variable of architecture comparison. Its hard content concerns causal geometry: paths, cuts, exact symmetries, typed composition, and robust resource bounds. Its central empirical content is the architecture- and training-conditioned distribution Π^Λ over implementation organizations. Its inverse content is a disciplined recovery problem. It does not infer semantic labels from architecture, identify one world per model, privilege an interpretability basis, or equate task ground truth with mechanism ground truth.

11.2 Typed equivalence reporting

Cross-model “sameness” is not a single relation. For a pair of systems define the *mechanism conservation profile*

$$\mathbf{C}(A, B) = (c_{\text{beh}}, c_{\text{alg}}, c_{\text{obj}}, c_{\text{trk}}, c_{\text{coord}}, c_{\text{act}}),$$

where each coordinate is a separately estimated equivalence claim with its own scope, tolerance, and uncertainty. Figure 13 shows the six claim types. They are not a hierarchy or lattice: two systems may conserve an algorithm while changing carriers, or align coordinates while implementing different algorithms. Appendix B gives the evidence requirements for each coordinate.

Physical specificity often demands stronger evidence, but the levels are not nested in all settings. For example, two systems can have aligned coordinates without implementing the same algorithm, or implement the same algorithm through non-transportable tracker families. Every comparison must state the level, gauge, scope, and tolerance.

11.3 Why method relativity is not relativism

Method and frame indexing does not imply that all stories are equally valid. It makes disagreements testable. Candidate worlds must predict data not used to fit or select them. Increasing the intervention/readout family can refine or destroy an equivalence class. Complexity constraints prevent arbitrary nonlinear alignments from manufacturing any desired abstraction (Sutter et al., 2025). Method disagreement is evidence about target mismatch or underdetermination, not permission to choose the most appealing story.

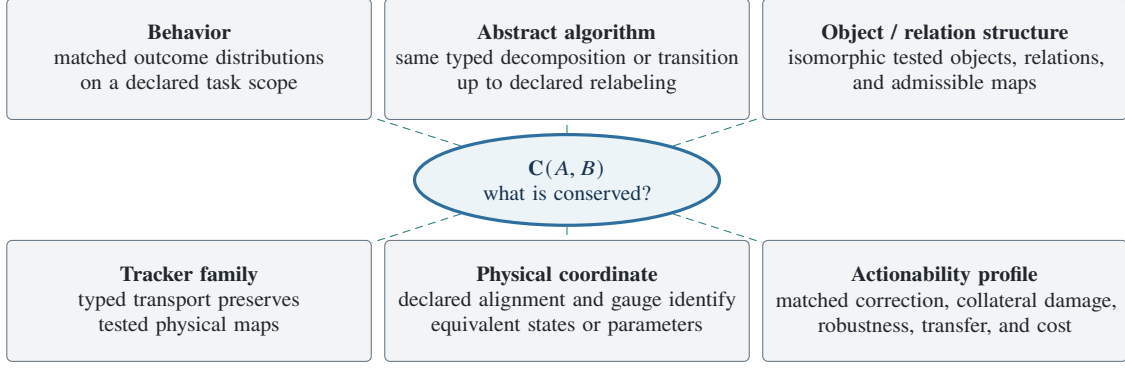


Figure 13: The mechanism conservation profile. Cross-model sameness decomposes into six independently tested coordinates. The dashed spokes associate each claim with the comparison; they deliberately do not form an implication chain.

11.4 Constructive closure

Readability, localization, and actionability are separate coordinates, but a strong mechanism program aims for a capability graph:

$$\text{predict held-out intervention} \longrightarrow \left\{ \begin{array}{l} \text{selective disable/rescue,} \\ \text{withheld composition,} \\ \text{typed transport,} \\ \text{reconstruction under shift.} \end{array} \right.$$

The arrows are research goals, not logical implications. A claim never inherits a downstream license without the corresponding experiment.

12 Limitations and epistemic status

The framework is not yet a calibrated predictive theory. It lacks results from the integrated empirical programme, a universal mechanism-class representation, and a validated estimator for continuous world families. The reachability, packing, and quotient ingredients are standard; the carrier-cut theorem is an elementary structural separation whose scientific value depends on whether its predicted redistribution of mechanism and failure is observed in trained populations. All formal statements depend on declared structural equations, readers, tolerances, precision/noise, and graph granularity. Mechanistic inductive bias is therefore a proposed target of estimation, not a measured universal law. Selective evidence synthesis is vulnerable to publication and recency bias; many 2026 sources are preprints frozen at the version available on July 27, 2026. The evidence atlas improves traceability but does not provide dual coding, exhaustive database search, or risk-of-bias assessment.

Architecture families contain heterogeneous implementations. Mechanism classes can remain non-identifiable even after extensive intervention, and MDL partitions depend on the coding language. Natural-donor interventions can still violate support, and reconstruction may exploit handles not used natively. The proposed Phase I is intentionally narrow; success would not establish the GNN, MoE, diffusion, multimodal, or embodied extensions in Appendix C.

13 Conclusion

Neural architecture research has a missing dependent variable. Expressivity, optimization, efficiency, and behavioral generalization do not determine how a trained model organizes computation. This paper names the corresponding second-order bias:

$$\Pi_{S_0, \mathcal{E}}^\Lambda = \mathcal{L}\left([\mathcal{W}_{\theta, z}^{\text{impl}}]_\Lambda\right).$$

Architecture supplies a causal syntax; training and context select among implementation organizations. The carrier-cut separation makes one consequence explicit: architectures can relocate computational pressure. A fixed robust cut pays exponentially growing distinguishability cost for independently queryable bindings, whereas extensive addressable carriers grow memory with the records and move difficulty toward selection and consumption. More generally, an architecture edit may change the mechanism and its failure interface before it changes accuracy.

The programme has a decisive acceptance criterion. Syntax must improve out-of-sample prediction of mechanism fingerprints beyond task, training, resources, and behavior. Controlled syntax edits must move those fingerprints in preregistered directions. Recovered organizations must predict evidence not used to construct them: interventions, compositions, transports, or reconstructions. If these tests fail, mechanistic inductive bias adds no scientific variable beyond ordinary behavioral inductive bias. If they succeed, the forward, inverse, and edit problems open a sequence of further layers: mechanism formation during training, switching under context, conservation across models, and constructive architecture design. The macroscopic explanandum is already present as organized model behavior and capability; what is missing is a lawful intermediate theory of why those capabilities arise, generalize, and fail. The present paper is the smallest closed and falsifiable *computational-mesoscale* layer of that programme, not its endpoint. A Hilbert-sixth-problem-style continuation would next require structural-semantic and grounding layers that are held to the same standards of causal identification, composition, transport, and constructive closure. Only if those layers can also be landed could the resulting stack serve as a common formal spine for model interpretability and alignment. The larger agenda is to turn architecture science from a theory of possible functions into a theory of possible and probable *internal computational worlds*: how they form, which structures recur across physical implementations, how they break, how they acquire grounded significance, and how they can be deliberately rebuilt in a new model or body. Nothing in the present framework closes those questions or axiomatizes intelligence; it makes the first layer precise enough to be continued rather than merely announced.

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A Complete Phase I empirical protocol

This appendix makes the empirical programme operational while keeping it subordinate to the theoretical claim. Phase I uses three tasks, four core architecture contrasts, three evidence regimes, and three method families. GNN, MoE, diffusion, multimodal, and embodied extensions enter only after the core estimators and failure criteria are calibrated.

A.1 Tasks and controlled factors

1. **Orbit transport.** The same abstract object appears under a known position or group action. The primary factors are seen/unseen group element, coordinate encoding, boundary condition, and parameter tying.
2. **Pointer lookup.** A query selects one of several matched records. Absolute position, relative distance, context length, candidate count, candidate similarity, and position syntax are varied factorially rather than collapsed into “retrieval difficulty.”
3. **Held-out composition.** Primitive typed maps $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are trained, while selected instances of $g \circ f$ are withheld. Primitive success is separated from physical compositional closure.

Task encodings are randomized across coordinate systems and counterbalanced so that a grid, sequence, or graph presentation does not trivially reveal the intended mechanism. Encoding is an explicit factor, not hidden preprocessing.

A.2 Architecture contrasts and resources

Core contrasts are:

1. untied versus parameter-tied MLP controls;
2. symmetry-preserving versus symmetry-breaking CNNs, with boundary and dilation edits;
3. RNN, linear SSM, and selective SSM variants with state-capacity, selectivity, and short-convolution edits;
4. Transformer variants with preregistered attention mask and position syntax.

Two analyses are reported. A resource-matched design declares one primary budget and reports all unmatched resources; an architecture-optimized analysis compares Pareto frontiers. Parameters, FLOPs, memory state, active compute, training time, and wall time are never claimed to be simultaneously matched.

A.3 Three evidence regimes

1. **Planted-mechanism organisms.** Physical variables, typed trackers, redundancy, repair pathways, and intervention ground truth are constructed. Recovery is scored against mechanism ground truth up to a declared gauge.
2. **Trained synthetic models.** Only the task generator is known. Internal mechanism is unknown. Scoring uses held-out interventional prediction, MDL/compression under a declared language, composition, transport, and selective reconstruction. Task objects are not scored as “true internal objects.”
3. **Natural models.** Claims are limited to predictive validity, typed equivalence, and actionability on declared distributions. No complete mechanism ground truth or total ontology recovery is asserted.

A.4 Methods, data roles, and search budgets

Phase I uses three method families: a controlled decoder/probe, natural-donor causal intervention with interaction-aware coalitions, and reconstruction/transplant. Each receives a transparent compute, supervision, intervention, and search budget appropriate to its operation; equal access is not assumed.

The four data roles are disjoint:

$$\begin{aligned} \mathcal{D}_{\text{fit}} &\rightarrow \text{fit sensors and alignments,} & \mathcal{D}_{\text{sel}} &\rightarrow \text{select candidates and hyperparameters,} \\ \mathcal{D}_{\text{eval}} &\rightarrow \text{estimate preregistered effects,} & \mathcal{D}_{\text{hold}} &\rightarrow \text{test structural generalization.} \end{aligned}$$

Automated search trajectories, donor optimization, prompt selection, stopping decisions, and candidate partitions belong to fitting or selection. Using them as confirmatory evidence requires selective-inference correction or a fresh holdout.

A.5 Primary tests

The central population target is whether syntax improves out-of-sample prediction of a mechanism class or fingerprint:

$$\begin{aligned} \mathcal{M}_0 : \quad y &\sim p(y \mid \mathcal{T}, \mathcal{E}, \text{Res}, \text{Behavior}), \\ \mathcal{M}_1 : \quad y &\sim p(y \mid \mathcal{T}, \mathcal{E}, \text{Res}, \text{Behavior}, S_0). \end{aligned}$$

The primary comparison is held-out proper-score improvement on held-out architecture edits and task encodings. In planted organisms y may be a mechanism class. In trained models it is a preregistered fingerprint vector or a complexity-controlled partition selected without holdout access.

Table 8: Phase I confirmatory skeleton. Numerical margins are pilot-dependent and must be frozen before the confirmatory run.

Question	Primary estimand	Required preregistration
Does syntax predict mechanism?	Held-out proper-score difference $\mathcal{M}_1 - \mathcal{M}_0$	Minimum relevant improvement, folds, hierarchy, seeds, stopping
Does tying improve transport?	Paired transport-defect difference	Equivalence/superiority margin, group law, donor support
Does a cut fail under load?	Capacity profile $\mathbf{I}(K)$ and packing/MI bound	Noise, precision, margin, K grid, reader class
Does address compete?	Position-matched effects of count and similarity	Factorial contrasts, position family, length, multiplicity
Do primitives compose?	Withheld Δ_o	Composite family, type map, nulls, effect margin
Does a syntax edit move the world distribution?	Paired or population edit effect on $\phi(\mathcal{W})$	Coupling assumptions, operational-use check, direction, margin

Table 9: Equivalence-reporting types and minimum discriminating evidence. The rows are not a total order.

Declared type	Equality/equivalence asserted	Minimum discriminating evidence
Behavior	Matched outcome distribution on a declared task scope	Held-out behavioral equivalence test with uncertainty
Abstract algorithm	Same typed input–output decomposition or state transition up to declared relabeling	Competing algorithm models and held-out intermediate counterfactuals
Object/relation structure	Isomorphic recovered carrier of objects, relations, and admissible maps	Natural-donor interchange, alternative partitions, and quotient defects
Tracker family	Typed partial transport between physical mechanism families preserves tested maps	Cross-position/model transport plus mechanism-level interventions
Physical coordinate	A declared alignment identifies equivalent states or parameters	Gauge specification, identifiability contract, and held-out coordinate transport
Actionability profile	Comparable correction, collateral damage, robustness, transfer, and cost	Matched constructive intervention under distribution shift

Power is simulated for the minimum architecture-aware score improvement, equivalence margins, edit effects, and key interactions. Confirmatory analyses freeze alpha, power, censoring, multiple-testing correction, and immutable stopping. A candidate class added after holdout inspection is exploratory and requires a new holdout.

A.6 Acceptance and rejection

The central claim is supported only if:

1. syntax improves held-out mechanism prediction beyond task, training, resource, behavior, and encoding baselines;
2. operationally used syntax edits move fingerprints in preregistered directions;
3. recovered candidates predict structural holdouts not used for fitting or selection;
4. effects replicate across seeds and at least one new task encoding.

It is rejected on the tested scope if architecture-aware prediction remains within the equivalence margin of the baseline and syntax edits fail to move mechanisms despite verified operational use.

B Detailed equivalence evidence requirements

Physical specificity often demands stronger evidence, but the types are not nested in all settings. Two systems can have aligned coordinates without implementing the same algorithm, or implement the same algorithm through non-transportable tracker families. Every comparison states type, gauge, scope, tolerance, and uncovered mass.

C Scoped architecture extensions

C.1 Message-passing GNNs

For a specific message-passing class with declared initial features, L layers expose at most the L -hop neighborhood, and expressivity can be bounded by the corresponding Weisfeiler-Lehman test (Xu et al., 2018). Node IDs, positional encodings, higher-order states, rewiring, or a global token change the class. Finite width does not imply finite

distinguishability without the robust assumptions of [proposition 15](#).

Over-squashing, oversmoothing, WL ambiguity, and optimization failure are separate targets. Over-squashing is a cut/information-pressure hypothesis; oversmoothing is contraction or homogenization; WL ambiguity is an expressivity equivalence; optimization is neither ([Alon and Yahav, 2021](#); [Jamadandi et al., 2024](#); [Wu et al., 2023](#)). Input graph topology is not automatically the model’s causal information topology. Rewiring/global edges are decisive edits only if their operational use is verified.

C.2 Sparse mixtures of experts

For N experts, top- k routing, and L layers, expert identities, active sets, weighted mixtures, and cross-layer handoffs are distinct objects. The number of active-set sequences is at most $\binom{N}{k}^L$; individual branch combinations can scale as k^L within a path. Neither expression is a semantic module count. Dropping is implementation-dependent: dropless, expert-choice, rerouting, and capacity-limited systems have different cuts ([Shazeer et al., 2017](#); [Fedus et al., 2022](#); [Xue et al., 2024](#)).

The candidate tracker is

hidden geometry $\rightarrow$ router decision $\rightarrow$ weighted expert set/path $\rightarrow$ consumer.

Expert identity alone is not a skill label. Discrete routing adaptivity is not placed on the same scalar as continuous attention adaptivity. The extension should compare expert, active-set, router-logit, full-path, and consumer-conditioned predictors with separate estimands.

C.3 Diffusion models and DiTs

Four notions must be separated.

1. A network $v_\theta(x, t, c)$ is non-autonomous when the update map depends explicitly on time t .
2. A sampler can be non-autonomous through a time-dependent noise/solver schedule even if the network omits t .
3. An autonomous map applied to changing state distributions is distributionally nonstationary, not mathematically non-autonomous.
4. Shared weights can yield different effective behavior across state regimes without changing the underlying autonomous map.

Time-embedding redundancy in some regimes is therefore a boundary on, not evidence for, universal explicit-clock use ([Chavez, 2026](#)).

Layer-by-time features and externally useful diffusion descriptors show recoverable dynamics, not native object identity ([Luo et al., 2023](#); [Tang et al., 2023](#)). A transport defect across timesteps must separate alignment error, sampling randomness, and solver error before being called path dependence. A stable world family uses the cross-input coherence criteria of [section 2.5](#).

A naive “CNN versus ViT” by “supervised versus diffusion” table is not a matched factorial, because U-Net and DiT differ in skips, conditioning, normalization, scale, and cross-attention. The appropriate design edits one component at a time or builds matched denoisers with the same conditioning and solver. DiT-specific evidence about interface-dependent circuits is currently scoped to controlled settings ([Peebles and Xie, 2023](#); [Wang et al., 2026](#)).

C.4 External validity

Multimodal, embodied, speech, time-series, and automated-interpretability systems are future validity domains, not part of the core evidence. Closed-loop agents additionally require environment dynamics, feedback policies, safety constraints, and intervention-induced distribution shift in $\mathfrak{F}$ and $\mathfrak{C}_{\text{do}}$. Automated search additionally requires selective-inference correction because the tool adaptively chooses hypotheses, prompts, donors, and stopping times ([Shaham et al., 2024](#)).

D Preregistration minimum

Before a confirmatory run, freeze:

1. task generators, encoding randomization, data splits, software, model versions, and seeds;
2. syntax variants and the primary resource budget;
3. method classes, search budgets, donor support, and stopping rules;
4. mechanism equivalence levels and candidate partitions;
5. all estimands, effect/equivalence margins, nulls, power simulations, multiplicity correction, and censoring rules;
6. architecture-edit predictions and contradiction categories;
7. held-out interventions, composites, contexts, and transport transformations;

8. a signed timestamp or immutable artifact hash.

E Evidence-file schema

The source package contains:

- `evidence_atlas.csv`: one row per cited source-claim pair with evidence axes, status, version date, scope, and limitation;
- `claim_to_source.csv`: manuscript claim identifiers and supporting/limiting sources;
- `search_protocol.md`: the non-systematic selective-synthesis protocol and freeze date;
- `exclusion_log.csv`: sources considered during revision but excluded from evidential support or moved to extensions.

These files support auditability without claiming exhaustive retrieval or dual-coded systematic review.